ENVS 193DS Final

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GitHub Repo

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # organizing files
library(rstatix) # statistical relationships
library(MuMIn) # modelling
library(DHARMa) # residual diagnostic simulation
library(ggeffects) # model predictions
library(ggplot2) # visualizations

# read in data
nests <- read_csv(here("data", "nest_data_final.csv"))
# read in and make object for nest data
dinner <- read_csv(here("data", "Dinner_Data.csv"))
# read in and make object for personal data
```

Problem 1. Research Writing

a. Transparent Statistical Methods

In part 1, the researchers used a Generalized Linear Model (GLM) to reach their conclusion and compute that p-value, since the claim answers the central question of how distance to water edge predicts the probability of American Avocet microhabitat use. Additionally, since the variables are binary (use of microhabitat or no use) and continous (distance from water edge) are non-parametric. Furthermore, since the response variable is binary we know the test was a logistic regression, a specific type of generalized linear regression. In part 2, they used a chi-square test because they are examining the relationship between two categorical variables, bird species and habitat type.

b. Suggestions for rewriting

Part 1.

We found that American Avocets do not use microhabitats in a random fashion, but rather with respect to distance to water edge. In this sense, distance to a water edge acts as a predictor for the likelihood American Avocet use a microhabitat.

Part 2.

When looking at the distribution of American Avocet, Forster's Tern, and Black-necked Stilt, we found that each species was not spread randomly among habitats. Rather, each species presence differed based on habitat. This conclusion takes into consideration managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat as habitats.

c. Further Analysis

Another variable that could influence the probability of American Avocet microhabitat use is the predator abundance, which could be count data or categorical with pre determined levels from low presence to high presence of predators. This variable will influence American Avocet habitat use because this species might avoid areas with high concentrations of predators as they pose a risk for injury, mortality, and reduce reproductive success.

Problem 2. Data Analysis

a. Clean and Wrangle

```
# create new object, nest_clean
nests_clean <- nests |>
  # keep only height, bird species, and ant species
  select(height_cm, case_control, ant_species) |>
  # filter to include on C. mimosae, C. sjostedti, and C. nigriceps
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create new column for scientific species names
  mutate(scientific_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti", #C. sjostedti
    "RRB" ~ "Crematogaster mimosae", # C. mimosae
    "BBR" ~ "Crematogaster nigriceps" # C. nigriceps
  )) |>
  # order the species
  mutate(scientific_name =
           as_factor(scientific_name),
         scientific_name = fct_relevel(scientific_name,
                                       "Crematogaster sjostedti",
                                       "Crematogaster mimosae",
                                       "Crematogaster nigriceps"))
```

```r
# display new structure of the data
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```r
# display 10 random rows from data
slice_sample(nests_clean, n =10)
```

```
# A tibble: 10 x 4
   height_cm case_control ant_species scientific_name
       <dbl>        <dbl> <chr>       <fct>
 1      183             0 RRB         Crematogaster mimosae
 2      232.            1 RRB         Crematogaster mimosae
 3      244.            0 RRB         Crematogaster mimosae
 4      298.            0 RRB         Crematogaster mimosae
 5      238.            1 RRB         Crematogaster mimosae
 6      168.            0 RRB         Crematogaster mimosae
 7       93             0 BBR         Crematogaster nigriceps
 8       65             0 BBR         Crematogaster nigriceps
 9      174.            0 BBR         Crematogaster nigriceps
10      286             0 AB          Crematogaster sjostedti
```

b. Response Variable

The 1s and 0s in the case_control column indicates whether or not a tree contained a nest. If a data point has a 1 this indicates the tree contains a nest, whereas a 0 indicates the tree did not contain a nest and was “available” for comparison.

c. Purpose of Study

Tree Height

Tree height is a numerical, continuous variable measured in cm. This variable can influence the probability of nest occurrence because as tree height increases

Acacia Ant Species

This is where I will describe the variable

d. Table of Models

Model number	Ant Species	Tree Height	Model description
0	X	X	All predictors (No Interaction term, +)
1	X	X	All predictors (Interaction term, *)

e. Run the Models

```
# model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ scientific_name + height_cm,
                   # no interaction model variables
                   data = nests_clean, # use clean dataset
                   family = binomial) # set to binomial

# model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ scientific_name * height_cm,
                   # interaction model variables
                   data = nests_clean, # use clean data set
                   family = binomial) # set to binomial
```

f. Select the Best Model

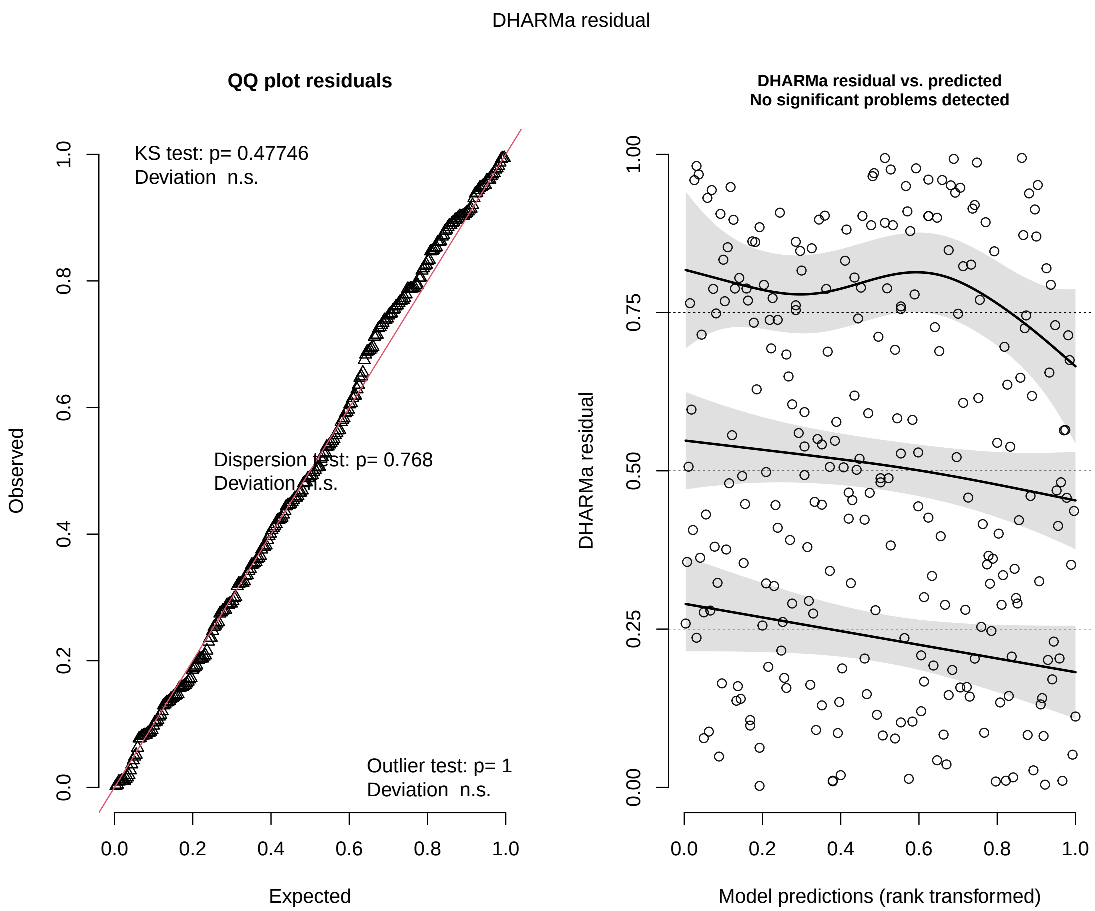
```
# calculate AIC
AICc(
  nest_model0,
  nest_model1
) |>
  arrange(AICc)
```

```
            df     AICc
nest_model1  6 238.1236
nest_model0  4 258.8940
```

The best model for predicting the probability of bird nest presence as determined by Akaike's Information Criterion (AIC) is the model of interaction between tree height (cm) and ant species, demonstrated by the lower AIC value.

g. Check the Diagnostics

```
# check diagnostic with DHARMa
plot(
  simulateResiduals(nest_model1)
)
```



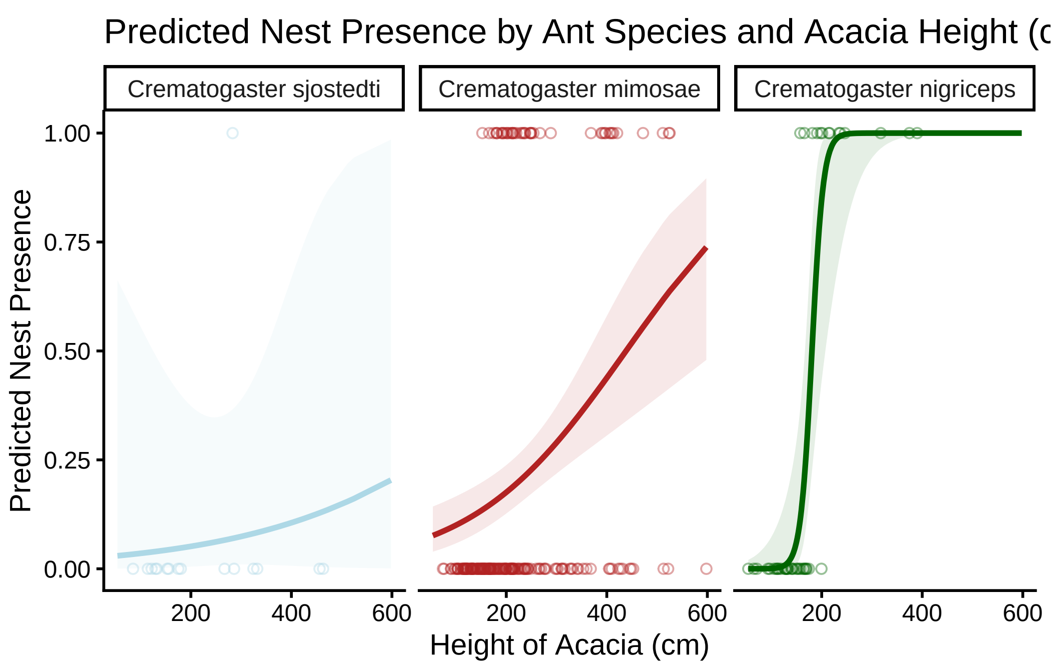
h. Visualize the Model Predictions

```
# create preds object
preds <- ggpredict(
  nest_model1, # model object 1
  terms = c("height_cm[all]", "scientific_name")
  # predictors height and scientific name
  )

# add scientific_name column for wrap in visualization
preds$scientific_name <- preds$group
```

```
# base layer: ggplot
ggplot(data = nests_clean,
       mapping = aes(x = height_cm,
                     y = case_control,
                     color = as.factor(scientific_name))) +
  # relabelling x- and y-axis
  labs(x = "Height of Acacia (cm)", # x-axis height
       y = "Predicted Nest Presence", # y-axis nest probability
       color = "scientific_name", # color by ant species
       title =
         "Predicted Nest Presence by Ant Species and Acacia Height (cm)")+
  # add tile
  # manually change ant colors
  scale_color_manual(values = c("Crematogaster sjostedti" = "lightblue",
                                "Crematogaster mimosae" = "firebrick",
                                "Crematogaster nigriceps" = "darkgreen")) +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "lightblue",
                                "Crematogaster mimosae" = "firebrick",
                                "Crematogaster nigriceps" = "darkgreen")) +
  # change theme from default
  theme_classic() +
  # remove gridlines and background
  theme(panel.background = element_rect()) +
  # remove legend
  theme(legend.position = "none") +
  # first layer: geom points, showing underlying data
  geom_point(mapping = aes(x = height_cm, # x-axis: aciacia height
                           y = case_control, # y-axis: nest occurence
                           color = scientific_name), # set to selected colors
             shape = 21, # change shape
```

```
            alpha = 0.4,# make slightly transparent
            inherit.aes = FALSE) + # ignore global aes for colors
# second layer: geom line, showing predictions
geom_line(data = preds, # data from model 1 prediction
          mapping = aes(x = x, # x-axis height of acacia
                        y = predicted, # y axis: prediction probability
                        color = group), # color by ant
          linewidth = 1) +
# third layer: ribbon (confidence interval)
geom_ribbon(data = preds, # data from model 1 prediction
            mapping = aes(x = x, # x-axis height of acacia
                          ymin = conf.low, # set low value
                          ymax = conf.high, # set high value
                          fill = group), # fill by ant species
            inherit.aes = FALSE, # ignore global aes
            alpha = 0.1) + # make transparent
# make new panel for each ant species
facet_wrap(~ scientific_name)
```



i. Write a Caption for your Figure

Figure 1. Probability of bird nest occurrence differs depending on the ant species present and acacia tree height Each figure separates the data, the prediction of nest occurrence given tree height (cm), and the 95% confidence interval based on ant species. The blue figure depicts data associated with the Crematogaster sjostedti, the red figure depicts data associated with Crematogaster mimosae, and the green figure represents data associated with Crematofaster nigriceps. Circles represent individual observations of nest occruences in each scenario, the lines represent model predictions of the porbability of nest occurences given tree height (cm) and ant species, and the ribbon represents the 95% confidence interval. Data originally from, Lujan, E., et al. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculate Model Predictions